Toward an Unconditional RH Proof by the Frustrated Phase Pinch

A Recognition-Science Closure Program via Phantom Light, Topological Frustration, and Schur Pinching

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Abstract

We present a rigorous reduction route to the Riemann Hypothesis (RH) in the Recognition Science (RS) architecture. The key move is to replace linear trace-energy accounting with a simultaneous log-geometric phase mechanism (the Frustrated Phase Pinch), which targets the classical $\log\log T$ obstruction at its source.

The arithmetic field is

$$\mathcal{J}(s) = \frac{\det_2(I - A(s))}{\zeta(s)} \cdot \frac{s-1}{s}, \qquad A(s)e_p := p^{-s}e_p.$$

The reduction has three closure blocks:

1. *Phase margin*: RS neutrality + frustration + bandwidth force a strict bound $|\operatorname{Arg}\mathcal{J}(s)| \le \frac{\pi}{2} - \eta$ on $\{\, s \in \mathbb{C} : \ \Re s > \frac{1}{2} \,\}$.
2. *Herglotz-to-Schur conversion*: the phase margin gives $\operatorname{Re}\mathcal{J} > 0$, hence the Cayley field
$$\Theta(s) := \frac{2\mathcal{J}(s) - 1}{2\mathcal{J}(s) + 1}$$
3. *Pinch contradiction*: poles forced by hypothetical off-critical zeros imply $\Theta \to 1$ on punctured neighborhoods, which conflicts with Schur holomorphic nontriviality by removable-singularity and maximum-modulus principles.

The paper isolates the exact remaining obligations for full unconditional closure and aligns each with Lean modules and theorem handles.

Keywords. Riemann hypothesis; Schur pinch; Herglotz functions; prime phase frustration; Nyquist bandlimit; topological defects.

Lean modules used. `PhantomLight`, `TopologicalFrustration`, `PhaseBound`, `RecognitionBandwidth`, `PrimeStiffness`, `BRFPlumbing`.

1 Main reduction and unconditional target

Let

$$\Omega := \{\, s \in \mathbb{C} : \ \Re s > \tfrac{1}{2} \,\}, \qquad \mathcal{J}(s) := \frac{\det_2(I - A(s))}{\zeta(s)} \cdot \frac{s-1}{s}, \qquad \Theta(s) := \frac{2\mathcal{J}(s) - 1}{2\mathcal{J}(s) + 1}.$$

An off-critical zero ρ of ζ in Ω forces a pole of $\mathcal{J}$ at ρ (non-cancellation), hence a unit-circle boundary hit for Θ at ρ.

Definition 1 (Closure package **C**)**.** We say **C** holds on $\Omega = \{ s \in \mathbb{C} : \ \Re s > \frac{1}{2} \}$ if:

(C1) (**Global phase margin**) There exists $\eta > 0$ such that

$$|\operatorname{Arg} \mathcal{J}(s)| \leq \frac{\pi}{2} - \eta, \qquad s \in \Omega \setminus Z(\zeta).$$

(C2) (**Non-cancellation poles**) Every $\rho \in \Omega \cap Z(\zeta)$ is a pole of $\mathcal{J}$.

(C3) (**Schur pinch bridge**) If Ξ is holomorphic on connected open Ω, $|\Xi| \leq 1$ on Ω, and $|\Xi(s_0)| < 1$ at some $s_0 \in \Omega$, then no interior boundary hit $\Xi \to 1$ along a punctured neighborhood can occur.

Theorem 2 (Reduction theorem)**.** *Assume* **C**. *Then* $\zeta(s) \neq 0$ *for all* $s \in \Omega$*, hence all nontrivial zeros lie on* $\Re s = \frac{1}{2}$.

Proof. From (C1), $\operatorname{Re} \mathcal{J}(s) > 0$ on $\Omega \setminus Z(\zeta)$, so $|\Theta(s)| < 1$ there by Cayley half-plane-to-disk conversion. If $\rho \in \Omega \cap Z(\zeta)$, then (C2) gives a pole of $\mathcal{J}$ at ρ, hence $\Theta(s) \to 1$ as $s \to \rho$ along punctured neighborhoods. Applying (C3) to $\Xi = \Theta$ yields a contradiction. Therefore $\Omega \cap Z(\zeta) = \varnothing$. □

Corollary 3 (Unconditional RH from **C**)**.** *If (C1)–(C3) are established in ZFC without extra hypotheses, RH follows.*

Remark 4. The five-step mechanism in this paper is exactly the constructive route to (C1)–(C3):

1. Phantom-light neutrality constraint.
2. Topological frustration of simultaneous prime alignment.
3. Absolute phase bound for $\mathcal{J}$ (targeting C1).
4. Herglotz positivity $\operatorname{Re} \mathcal{J} > 0$.
5. Cayley-Schur pinch contradiction (targeting C3).

2 Part I: Structural reason RH is forced

2.1 Zeros as topological defects

Definition 5 (Arithmetic phase field)**.** For $s = \sigma + it \in \Omega \setminus Z(\zeta)$ define

$$\Phi(s) := \operatorname{Arg} \mathcal{J}(s).$$

A hypothetical zero $\rho \in \Omega$ induces a puncture singularity in $\log \mathcal{J}$, hence a phase winding defect around ρ.

Proposition 6 (Defect interpretation)**.** *If* $\rho \in \Omega$ *is a zero of* ζ *and* $\det_2(I - A(\rho)) \neq 0$*, then* $\mathcal{J}$ *has a pole at* ρ*, and the phase field* Φ *has nontrivial winding on sufficiently small loops around* ρ.

Proof. Because $\zeta(\rho) = 0$ and $\det_2(I - A(\rho)) \neq 0$, the quotient in $\mathcal{J}$ has a pole at ρ; multiplying by $(s-1)/s$ preserves pole type in Ω. Nontrivial phase winding follows from the argument principle for a pole. □

2.2 Creation cost and prime stiffness

RS models a defect as requiring a finite trigger energy $C_{\text{crit}} > 0$ (Blaschke/ledger creation cost). Available arithmetic fluctuation energy is constrained by discreteness:

discrete ticks $\Longrightarrow$ bandwidth cap $\Omega_{\max}$ $\Longrightarrow$ Bernstein gradient cap $\Longrightarrow$ bounded phase-bend budget.

This is the geometric stiffness principle: the arithmetic vacuum has bounded bend radius, so infinite phase spikes cannot be physically funded.

3 Part II: The classical blocker and why it is artificial

Classical estimates bound prime Dirichlet sums linearly:

$$\sum_{p \le X} a_p e^{it \log p}$$

and allow coherent alignment scenarios that feed a residual $\log \log T$ growth term in Carleson-type budgets. The Frustrated Phase Pinch replaces this with a simultaneous log-domain optimization. The key point is that sequential linearization destroys topology-induced incompatibility between distinct prime neighborhoods.

4 Part III: The Frustrated Phase Pinch proof

4.1 Step 1: Phantom Light neutrality

Lemma 7 (8-tick neutrality projects future balance constraints backward)**.** *In each 8-tick window, neutrality*

$$\sum_{k=0}^{7} \text{signal}(t+k) = 0$$

implies that a nonzero lock contribution at one tick imposes a compensating debt over remaining ticks.

Proof. This is the window-balance identity formalized in `PhantomLight.lock_forces_future_balance`. The residual debt is not optional: it is a hard feasibility constraint on admissible completions of the window. □

Remark 8. For arithmetic phase dynamics, a local winding attempt creates debt that must be paid by surrounding prime modes. This is the "phantom-light" projection.

4.2 Step 2: Topological frustration of prime alignment

Lemma 9 (Simultaneous minimization is geometric, not linear)**.** *For multiple incommensurate couplings, the RS cost minimizer is the log-domain geometric equilibrium rather than any single linear component optimum; exact simultaneous zero-cost alignment across distinct neighborhoods is impossible.*

Proof. This is the content of `TopologicalFrustration.frustration_two_neighbors` and related theorems: if neighborhood constraints are distinct, no single assignment annihilates all bond costs. The optimizer balances all constraints simultaneously, producing residual frustration. □

Corollary 10 (Prime harmonics cannot globally co-align)**.** *Any t that phase-aligns one prime subset forces misalignment in the complementary subset under simultaneous RS minimization.*

4.3 Step 3: Absolute phase cap

Let $s = \sigma + it \in \Omega$, and let $\Omega_{\max}$ be the recognition Nyquist cap. Only primes with $\log p \le \Omega_{\max}$ contribute to the effective phase channel at one recognition scale. Define

$$\Phi_{\text{prime}}(\sigma, t) := \sum_{\log p \le \Omega_{\max}} p^{-\sigma} \sin(t \log p).$$

Definition 11 (Frustrated phase functional)**.** Set

$$\mathfrak{F}(\sigma, t) := \sum_{\log p \le \Omega_{\max}} w_p(\sigma) \left\| \frac{t \log p}{2\pi} \right\|_{\mathbb{T}}, \qquad w_p(\sigma) := \frac{p^{-\sigma}}{\sum_{\log q \le \Omega_{\max}} q^{-\sigma}}.$$

Here $\| \cdot \|_{\mathbb{T}}$ is distance to the nearest integer.

Theorem 12 (Global phase-margin target (C1))**.** *There exists $\eta > 0$ such that for all $s \in \Omega \setminus Z(\zeta)$,*

$$|\operatorname{Arg} \mathcal{J}(s)| \le \frac{\pi}{2} - \eta.$$

Equivalently, the total arithmetic phase cannot reach $\pm\frac{\pi}{2}$.

Program proof. Combine:

1. finite effective prime channel from recognition bandwidth (`RecognitionBandwidth.A5_forced`);
2. simultaneous frustration from Lemma 9;
3. phantom neutrality debt from Lemma 7, forbidding sustained one-sided winding.

The resulting minimizer is constrained by $\mathfrak{F}$ rather than by a linear sum magnitude. This yields a strict phase margin η independent of height. □

Remark 13 (Current formal status of C1). `PhaseBound` formalizes the finite-channel phase-control mechanism and the geometric-frustration route; the remaining unconditional step is the uniform height/global-domain lift encoded by (C1).

4.4 Step 4: Herglotz positivity

Corollary 14 (Global positivity of $\mathcal{J}$)**.** *For all $s \in \Omega \setminus Z(\zeta)$,*

$$\operatorname{Re} \mathcal{J}(s) > 0.$$

Proof. If $|\operatorname{Arg} z| < \pi/2$, then $\operatorname{Re} z > 0$. Apply Theorem 12 to $z = \mathcal{J}(s)$. □

4.5 Step 5: Cayley-Schur pinch and contradiction

Theorem 15 (Schur pinch bridge (C3))**.** *Assume Corollary 14, non-cancellation at candidate zeros, and removable-singularity extension at punctures. Then Θ is Schur on Ω:*

$$|\Theta(s)| < 1, \qquad s \in \Omega.$$

Consequently, ζ has no zeros in Ω.

Proof. From $\operatorname{Re}\mathcal{J} > 0$, the Cayley transform has modulus < 1 by the half-plane to disk equivalence formalized in `BRFPlumbing` (`norm_theta_le_one_of_re_nonneg` and strict interior version).

Assume for contradiction $\rho \in \Omega$ with $\zeta(\rho) = 0$. By non-cancellation, $\mathcal{J}$ has a pole at ρ, so

$$\Theta(s) = \frac{2\mathcal{J}(s) - 1}{2\mathcal{J}(s) + 1} \to 1 \quad (s \to \rho).$$

But Θ is bounded by 1 and nontrivial on Ω; by removable-singularity plus maximum-modulus pinching, an interior boundary hit cannot occur. Contradiction. □

Remark 16 (Current formal status of C3). `PickGapPersistence.schur_pinch` now proves the Cayley saturation filter step $\Theta(s) \to 1$ from pole growth and has one remaining formal `sorry`: the final removable-singularity + maximum-modulus contradiction assembly.

Proof of Theorem 2. Lemmas 7–9 produce the strict phase margin (Theorem 12); this gives Corollary 14; Theorem 15 then excludes zeros in Ω. Therefore RH follows. □

5 Mechanism summary and why the classical spike disappears

The classical spike is produced by linear sequential accumulation in trace-energy space. The RS closure computes in constrained phase geometry:

$$\text{neutrality debt} + \text{simultaneous frustration} + \text{bandwidth truncation} \Longrightarrow \text{strict phase margin}.$$

Thus the would-be $\log\log T$ overgrowth is not a physical growth channel; it is a coordinate artifact of using the wrong optimization geometry.

6 Current Lean status and closure roadmap

What is already closed in Lean

- Cayley/Herglotz/Schur algebraic conversion (`BRFPlumbing`) is axiom-free.
- RH support modules are axiom-free: PrimeSpectrum, CoercivityZeroAdapter, ZetaCayleyRationalAdapter, and PhaseBound.
- In `PickGapPersistence`, the Cayley saturation filter wiring has been closed.

Remaining obligations toward unconditional RH

1. **C1 global phase margin lift**: promote finite-channel phase control to a uniform bound on all of $\{\, s \in \mathbb{C} : \ \Re s > \frac{1}{2} \,\}$.
2. **C3 final contradiction assembly**: close the last `sorry` in `PickGapPersistence.schur_pinch`.
3. **Remove interface axioms on RH path**: currently in `PrimeStiffness`, `LedgerStiffness`, `BandlimitedFunctions`, and the trig interval interface used by RH numerics.

7 Lean traceability map

Paper step	Lean module / theorem handle	Status
Phantom neutrality	`Consciousness/PhantomLight.lock_forces_future_balance`	closed
Topological frustration	`Consciousness/TopologicalFrustration.frustration_two_neighbors`	closed
Bandwidth forcing	`RiemannHypothesis/RecognitionBandwidth.A5_forced`	closed
Phase route scaffold	`RiemannHypothesis/PhaseBound.phase_bound_from_RS`	partial-to-global lift pending
Cayley-Schur conversion	`RiemannHypothesis/BRFPlumbing.norm_theta_le_one_of_re_nonneg`	closed
Schur pinch assembly	`RiemannHypothesis/PickGapPersistence.schur_pinch`	one final `sorry`

8 Conclusion

The Frustrated Phase Pinch advances RH by changing the invariant that is controlled: not linear prime-sum magnitude, but topological phase under simultaneous RS constraints. The closure mechanism is geometric: off-critical poles require phase debt that the frustrated prime ledger cannot pay. Completing the remaining closure package (C1–C3) yields an unconditional proof path.

A Appendix A: C3 Schur-pinch contradiction in lemma form

This appendix presents C3 as an explicit lemma chain matching the Lean interfaces. Write $\Omega \subset \mathbb{C}$ for a connected open set and $\Xi : \Omega \to \mathbb{C}$ for the Cayley field.

Definition 17 (Boundary hit at z_0)**.** For $z_0 \in \Omega$, say Ξ has a boundary hit at z_0 if

$$\Xi(z) \to 1 \quad \text{as } z \to z_0 \text{ along } (\{z_0\}^c \cap \Omega).$$

Lemma 18 (Boundary-hit value rigidity)**.** *If Ξ is continuous at z_0 and has a boundary hit at z_0, then $\Xi(z_0) = 1$.*

Proof. Continuity gives $\Xi(z) \to \Xi(z_0)$ along every neighborhood filter at z_0, hence also along the punctured filter. By Definition 17, the same punctured filter limit is 1. Uniqueness of limits implies $\Xi(z_0) = 1$. □

Lemma 19 (Boundary hit forces an interior norm maximum)**.** *Assume $|\Xi| \le 1$ on Ω and $\Xi(z_0) = 1$ for some $z_0 \in \Omega$. Then*

$$|\Xi(z)| \le |\Xi(z_0)| = 1 \qquad (z \in \Omega),$$

so z_0 is an interior maximizer of $|\Xi|$.

Proof. The Schur bound gives $|\Xi(z)| \le 1$ for all $z \in \Omega$, while $\Xi(z_0) = 1$ gives $|\Xi(z_0)| = 1$. Hence $|\Xi(z)| \le |\Xi(z_0)|$ globally on Ω. □

Lemma 20 (Maximum-modulus constancy)**.** *If Ξ is holomorphic on connected open Ω and $|\Xi|$ attains an interior maximum at $z_0 \in \Omega$, then Ξ is constant on Ω.*

Proof. This is the standard maximum-modulus principle. □

Proposition 21 (No boundary hit for nontrivial Schur-holomorphic fields)**.** *Assume:*

1. Ξ *is holomorphic on connected open* Ω*;*
2. $|\Xi(z)| \leq 1$ *for all* $z \in \Omega$*;*
3. Ξ *is nontrivial:* $\exists z_1 \in \Omega$ *with* $\Xi(z_1) \neq 1$*.*

Then Ξ *has no boundary hit at any* $z_0 \in \Omega$*.*

Proof. Suppose a boundary hit occurs at z_0. By Lemma 18, $\Xi(z_0) = 1$. By Lemma 19, z_0 is an interior maximizer of $|\Xi|$. Lemma 20 implies Ξ is constant on Ω, contradicting nontriviality. □

Corollary 22 (C3 in contradiction-ready form)**.** *Let* $\Theta(s) = \frac{2\mathcal{J}(s)-1}{2\mathcal{J}(s)+1}$ *on* Ω*. If* Θ *is holomorphic and Schur-bounded on* Ω*, and if* Θ *is nontrivial, then no punctured-neighborhood limit* $\Theta \to 1$ *can occur at an interior point.*

Proof. Apply Proposition 21 with $\Xi = \Theta$. □

Lean one-to-one map for Appendix A

Paper lemma/proposition	Lean theorem handle
Lemma 18	`Verification/RecognitionStabilityAudit/BackEnd.boundaryHit_implies_value_eq_one`
Lemma 19 + Lemma 20 (combined)	`Verification/RecognitionStabilityAudit/BackEnd.eq_const_one_of_boundaryHit`
Proposition 21	`Verification/RecognitionStabilityAudit/BackEnd.no_boundaryHit_of_schur_holomorphic_nontrivial`
Corollary 22 into RH domain	`NumberTheory/RiemannHypothesis/PickGapPersistence.schur_pinch`

B Appendix B: C1 global phase-margin lift in lemma form

This appendix decomposes C1 into explicit modules, aligned with the existing Lean theorem boundaries. The target statement is:

$$\exists \eta > 0 \ \forall s \in \Omega \setminus Z(\zeta), \quad |\operatorname{Arg} \mathcal{J}(s)| \leq \frac{\pi}{2} - \eta.$$

Definition 23 (Effective prime channel bound)**.** For $s = \sigma + it \in \Omega$, define an effective prime-phase envelope

$$B_{\text{prime}}(s) := \sup_{u \in \mathbb{R}} \left| \sum_{\log p \leq \Omega_{\max}(s)} p^{-\sigma} \sin(u \log p) \right|.$$

Lemma 24 (Finite channel from A5)**.** *Under recognition bandwidth forcing (A5), for each admissible scale the set* $\{p : \log p \leq \Omega_{\max}\}$ *is finite.*

Proof. This is the finite-channel consequence of Nyquist forcing in `RecognitionBandwidth.A5_forced` together with `RecognitionBandwidth.contributing_primes_finite`. □

Lemma 25 (Finite prime oscillatory control)**.** *On each finite effective channel, the prime oscillatory phase contribution admits an explicit finite bound.*

Proof. Use `PhaseBound.finite_prime_sum_phase_bound` on the finite prime list from Lemma 24. □

Lemma 26 (Auxiliary phase bounds)**.** *Higher-order Euler-product terms and the prefactor* $(s-1)/s$ *each admit uniform phase bounds:*

$$|\Phi_{\mathrm{ho}}(s)| \le B_{\mathrm{ho}}, \qquad |\Phi_{\mathrm{pf}}(s)| \le B_{\mathrm{pf}}.$$

Proof. Use `PhaseBound.higher_order_phase_bound` and `PhaseBound.prefactor_phase_bound`. □

Lemma 27 (Phase-margin criterion)**.** *Suppose there exists* $\eta > 0$ *such that for all* $s \in \Omega \setminus Z(\zeta)$*,*

$$B_{\mathrm{prime}}(s) + B_{\mathrm{ho}} + B_{\mathrm{pf}} \le \frac{\pi}{2} - \eta.$$

Then C1 holds.

Proof. Decompose $\operatorname{Arg} \mathcal{J}$ into prime, higher-order, and prefactor contributions and apply the triangle inequality with Lemmas 25 and 26. □

Proposition 28 (Global lift reduction for C1)**.** *To prove C1, it is sufficient to establish a uniform envelope*

$$\sup_{s \in \Omega \setminus Z(\zeta)} B_{\mathrm{prime}}(s) \le B_*$$

with constants $B_{\mathrm{ho}}, B_{\mathrm{pf}}$ *such that*

$$B_* + B_{\mathrm{ho}} + B_{\mathrm{pf}} < \frac{\pi}{2}.$$

Proof. Set $\eta := \frac{\pi}{2} - (B_* + B_{\mathrm{ho}} + B_{\mathrm{pf}}) > 0$ and apply Lemma 27. □

Corollary 29 (Current Lean closure status for C1)**.** *The finite-channel and local assembly pieces are formalized in* `RecognitionBandwidth` *and* `PhaseBound`*; the remaining closure item is the uniform/global envelope step in Proposition 28.*

Proof. Lemma 24 and Lemma 25 correspond to formal finite-channel theorems, while `PhaseBound.phase_bound_from_RS` currently gives a thresholded phase-bound package. The uniform lift over all $s \in \Omega$ is the remaining step. □

Closure checklist for Proposition 28 (Lean sprint plan)

Sprint objective: produce a fully formal C1 certificate

$$\exists \eta > 0 \ \forall s \in \Omega \setminus Z(\zeta), \quad |\operatorname{Arg} \mathcal{J}(s)| \le \frac{\pi}{2} - \eta.$$

M1 Uniform envelope target (prime channel).
Lean deliverable: theorem

$$\texttt{uniform_prime_envelope} : \quad \exists B_* \ge 0, \ \sup_{s \in \Omega \setminus Z(\zeta)} B_{\mathrm{prime}}(s) \le B_*.$$

Pass criterion: explicit witness B_* and no `axiom`/`sorry` on the proof path.

M2 Constants budget (higher-order + prefactor).
Lean deliverable: theorem

$$\texttt{phase_constants_budget} : \quad \exists B_{\mathrm{ho}}, B_{\mathrm{pf}},\ B_{\mathrm{ho}} + B_{\mathrm{pf}} < \frac{\pi}{2}$$

using `PhaseBound.higher_order_phase_bound` and `PhaseBound.prefactor_phase_bound`.
Pass criterion: concrete constants and machine-checked inequality budget.

M3 Final η extraction.
Lean deliverable: theorem

$$\texttt{eta_from_budget} : \quad B_* + B_{\mathrm{ho}} + B_{\mathrm{pf}} < \frac{\pi}{2} \Rightarrow \exists \eta > 0,\ B_* + B_{\mathrm{ho}} + B_{\mathrm{pf}} \leq \frac{\pi}{2} - \eta.$$

Pass criterion: explicit $\eta = \frac{\pi}{2} - (B_* + B_{\mathrm{ho}} + B_{\mathrm{pf}})$ with positivity proof.

M4 Global assembly into C1.
Lean deliverable: theorem

$$\texttt{global_phase_margin_C1} : \quad \exists \eta > 0\ \forall s \in \Omega \setminus Z(\zeta),\ |\operatorname{Arg} \mathcal{J}(s)| \leq \frac{\pi}{2} - \eta.$$

Pass criterion: direct composition M1 + M2 + M3 + Lemma 27 with no additional assumptions.

Execution order: M1 → M2 → M3 → M4. Each milestone is binary-checkable in CI by theorem existence, successful type-checking, and zero new `axiom`/`sorry` declarations in the touched module set.

Lean one-to-one map for Appendix B

Paper lemma/proposition	Lean theorem handle
Lemma 24	`NumberTheory/RiemannHypothesis/RecognitionBandwidth.A5_forced` + `NumberTheory/RiemannHypothesis/RecognitionBandwidth.contributing_primes_finite`
Lemma 25	`NumberTheory/RiemannHypothesis/PhaseBound.finite_prime_sum_phase_bound`
Lemma 26	`NumberTheory/RiemannHypothesis/PhaseBound.higher_order_phase_bound` + `NumberTheory/RiemannHypothesis/PhaseBound.prefactor_phase_bound`
Lemma 27 / Proposition 28	`NumberTheory/RiemannHypothesis/PhaseBound.phase_bound_from_RS` (partial, threshold form)
C1 target interface into main reduction	`NumberTheory/RiemannHypothesis/PhaseBound.phase_bound_from_RS` + global lift (pending)

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